

Seongkyu CHOI

M.S. Student, Gachon University

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EDUCATION

Gachon University, Seongnam, South Korea (Mar. 2024 - Present)

M.S. in Artificial Intelligence

- Department of AI & Software
- Graduate School, Master's Program
- Research Area: Computer Vision, Autonomous Driving, Off-Road, Edge AI Devices

Hansei University, Gunpo, South Korea (Mar. 2019 - Feb. 2021)

B.S. in Department of International Business

- Department of International Business

Shingu College, Seongnam, South Korea (Mar. 2015 - Feb. 2019)

Associate Degree in Business Administration

- Department of Global Business
- Transferred to Hansei University

RESEARCH INTERESTS

- Semantic segmentation in LiDAR point clouds and utilizing terrain information.
 - Computer Vision for Autonomous Driving and Robotics
 - Off-Road Semantic Segmentation under Ambiguous Boundaries and Noisy Supervision
 - Efficient On-Board AI Model Optimization for Edge Devices
 - Multi-Modal Perception using Camera, LiDAR, and Thermal Sensors
 - 3D Occupancy Prediction and Terrain Understanding

WORK EXPERIENCE

Research Experience

- Research Student, Vehicle Intelligence Perception Lab, Gachon University, Korea (Mar. 2024 ~ Present)
 - Developed perception AI models for Off-Road robots, autonomous driving and on-board edge AI systems.
 - Designed and validated semantic segmentation, traversability recognition and multi-task perception models using camera, LiDAR and edge devices.
 - Implemented on-board AI optimization using Jetson Orin, TensorRT, Docker, ROS2 and evaluated real-time performance using latency, FPS and memory efficiency
 - Validated perception models in real-world robotic and vehicle scenarios, focusing on robustness under complex outdoor conditions.

Work Experience

- Engineer, FNS, Korea (June 2021 ~ Mar. 2023)
 - Performed inspection, maintenance, troubleshooting, and recovery of equipment used in semiconductor clean processes.
 - Conducted pre-shipment and post-shipment equipment checks to verify operational stability and quality requirements.
 - Provided on-site technical support at customer semiconductor manufacturing sites.

- Analyzed recurring equipment issues and supported preventive actions to improve operational reliability.
- Developed problem analysis and troubleshooting skills through real-world industrial system operation and failure handling.

PROJECTS

Ongoing

- Development of Wide Semantic Map Building and Update Technology for Multi-Robot Systems in Unstructured Outdoor Environments (Apr. 2025 – Mar. 2027)
 - National R&D Project, led by Mobiltech
 - Funded by: Ministry of SMEs and Startups, Korea
 - Role: Off-road Perception Model Development
 - ◆ Developed semantic segmentation algorithms by integrating 3D point cloud and RGB information for unstructured outdoor environments.
 - ◆ Improved traversable area recognition by designing perception models for off-road robots operating in complex terrain.
 - ◆ Participated in the development of path similarity recognition and change detection methods using multi-scan sensor data.
 - ◆ Conducted performance evaluation and optimization of AI models for wide-area semantic map construction and update.
- Development of Intelligent Battlefield Situational Awareness AI Algorithms (Nov. 2024 – Sep. 2026)
 - Industry-commissioned Research Project
 - Funded by: Hyundai Rotem
 - Role: Data Curation and Augmentation
 - ◆ Collected battlefield-related object data, including tanks and armored vehicles, under diverse environmental conditions.
 - ◆ Performed data labeling and dataset construction for object recognition in battlefield situational awareness scenarios.
 - ◆ Designed data augmentation strategies to address limited training data and improve class diversity.
 - ◆ Built a data preprocessing and augmentation pipeline to improve model robustness and training efficiency.
- Development of an Automotive Software Platform for Software-Defined Vehicles (SDV) Integrated with an AI Framework Required for Intelligent Vehicles (Apr. 2024 – Dec. 2027)
 - National R&D Project, led by HL Mando / Mando
 - Funded by: Institute for Information & Communications Technology Planning & Evaluation (IITP), Korea
 - Role: On-board AI Model Optimization
 - ◆ Designed an on-board multi-task AI system for simultaneous detection, segmentation, and tracking in vehicle environments.
 - ◆ Applied lightweighting and TensorRT-based acceleration, and evaluated latency, FPS, and FPS/Memory(MB) for real-time edge deployment.
 - ◆ Integrated and tested the perception pipeline using Jetson Orin and ROS2 in real-vehicle driving scenarios.
 - ◆ Analyzed the trade-off between model accuracy, inference speed, and memory efficiency for stable on-board deployment.

Completed

- Development of a Network for Determining Traversable Regions by Terrain (Sep. 2024 – Feb. 2025)
 - Industry-Academic Collaborative Research Project

- Funded by: National IT Industry Promotion Agency (NIPA) with Mobiltech & GINT
- Role: Vision-based Perception Model Development and Validation
 - ◆ Developed a terrain-aware perception model for traversable region estimation in off-road agricultural environments.
 - ◆ Investigated vision-based autonomous driving technologies including obstacle detection, crop-row perception, and GNSS-Vision fusion systems.
 - ◆ Conducted experiments on object detection and semantic terrain understanding using camera and LiDAR sensor configurations.
 - ◆ Evaluated real-time deployment feasibility on NVIDIA Jetson embedded platforms for autonomous tractor applications.
 - ◆ Supported dataset analysis and validation for agricultural perception scenarios including obstacle-aware navigation and safety monitoring.
- Development of Camera-Based patrol cognitive intelligence for abnormal situation detection (Nov. 2024 – Mar. 2025)
 - Industry-commissioned Research Project
 - Funded by: Neubility
 - Role: Data Collection and Model Validation
 - ◆ Compared VLM models for abnormal situation understanding in patrol/security robot scenarios.
 - ◆ Evaluated edge-device applicability on Jetson Orin Nano using quantization, FPS, and VRAM usage.
 - ◆ Designed and validated a Descript-YOLO structure combining YOLO-based object detection and VLM-based situation description within a ROS2/Docker environment.
 - ◆ Refined the dataset by selecting representative abnormal-situation samples and adding missing classes such as smoke for robust anomaly detection.

PUBLICATIONS AND PRESENTATIONS

International Journal Articles

- Nahyeong Kim, **Seongkyu Choi**, Sun Choi, Yejun Lee, Youngjae Cheong, and Jhonghyun An, “GPT-4off: On-Board Traversability Probability Estimation for Off-Road Driving via GPT Knowledge Distillation,” Applied Sciences, vol. 15, no. 4, Art. no. 2130, Feb. 2025. Equal contribution by all authors.
- **Seongkyu Choi** and Jhonghyun An, "CSTR: Cross-Scale Token Refinement with Gated Structural Injection for Off-Road Segmentation,” Applied Sciences, vol. 16, no. 11, Art. no. 5238, May. 2026.

Manuscripts Under Review

- **Seongkyu Choi** and Jhonghyun An, “Enhancing Group Attention for Off-Road Semantic Segmentation via Transition-Aware Refinement,” Under review to IEEE Signal Processing Letters, under review, Apr. 2026.

RELATED COURSES

- Advanced Topics in AI Autonomous Driving A0
- Internet and Mobile Computing A0
- Fashion Big Data Analysis and Application A+
- Understanding of Advanced Artificial Intelligence A+
- Probabilistic Robotics A+
- Intelligent Vehicle Technology A+

AWARDS

- Outstanding Poster Paper Award, Korean Society of Next-generation Computing Conference, 2024
 - Paper: “Real-time Performance Optimization of AI Models Using Edge Computing”

Technical Skills

Languages

Python 
C 

Tools

DevOps: Docker, AWS, Jetson Orin, TensorRT
Robotics: ROS1, ROS2
Tools: Git, GitHub, VScode, Anaconda, Android Studio
OS: macOS, Windows, Linux